A Prompt Framework for Expert-Free LLM-Based Fully Automated Simple Power Analysis on Cryptosystems

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***Abstract*—Side-channel analysis is a powerful technique to extract secret data from cryptographic devices. However, this task heavily relies on experts and specialized tools, particularly in the case of simple power analysis (SPA). Meanwhile, large language models (LLMs) have demonstrated remarkable capabilities in assisting users with complex tasks, yet their potential for fully automated SPA remains unexplored. In this paper, we propose a novel prompt framework specifically designed for SPA tasks, enabling non-experts to perform end-to-end automated analysis by providing traces and prompts to an LLM in a single interaction. Our framework consists of six components: role, reinforce, input, task description, chain-of-thought, and expert strategies, with three expert strategies instantiated for SPA. To validate the framework's effectiveness and generalizability, we establish a dataset comprising seven sets of real power traces from various implementations of public-key cryptosystems, including RSA, ECC, and Kyber, as well as eighteen sets of simulated power traces that illustrate typical SPA leakage patterns. We compare against a representative unsupervised horizontal clustering attack, which achieves only 52.04% accuracy. In contrast, our framework enables GPT-4o and DeepSeek-V3.1 to achieve overall average accuracies of 98.02% and 97.83%, respectively. Furthermore, our framework reduces the analysis time by over 93.09% compared to manual analysis at comparable accuracy, enabling non-experts to complete SPA tasks in minutes rather than hours. Notably, this work represents the first successful application of LLMs to achieve fully automated SPA.**

***Index Terms*—Side-channel Analysis, Large Language Model, Public-key Algorithms, Kyber, Simple Power Analysis.**

I. Introduction

Recently, the swift advancement of AI technology has led to a remarkable surge in powerful large language models (LLMs). Leading global companies like OpenAI [1], Deepseek [2], and Google, along with numerous open-source contributors, have played a pivotal role in advancing the development of a wide range of LLMs. To date, LLMs have achieved significant success and found widely used in various domains, including code generation [3], vulnerability management [4], and anomaly detection [5]. Among the numerous LLMs, OpenAI's GPT series and the open-source DeepSeek models have attracted widespread attention for their exceptional performance across diverse tasks. Notably, GPT-4o demonstrates advanced multimodal capabilities and reasoning abilities, while DeepSeek-V3.1 showcases competitive performance with efficient architecture design.

Hardware and embedded systems security is essential, requiring cryptographic moudles to protect. Before entering the market, these products typically need experts and specialized tools to perform security tests. Among various security evaluation techniques, side-channel analysis plays a particularly critical role in assessing the resistance of cryptographic implementations to physical attacks. It enables attackers or evaluators to extract sensitive information by observing side-channel leakages such as power consumption, electromagnetic emissions, or timing information. In 1999, Kocher first proposed side-channel analysis for cryptosystems and successfully recovered the key using timing analysis and simple power analysis (SPA) [6]. Public-key cryptosystems such as RSA, ECC, and Kyber (standardized as ML-KEM in FIPS 203) are widely used for identity recognition and digital signatures. These algorithms demonstrate distinctive power consumption patterns, rendering them vulnerable to SPA [7]. Consequently, SPA has become a crucial step for evaluating the security of cryptographic devices, as outlined in ISO/IEC 17825 [8]. Researchers have developed several effective SPA techniques by combining machine learning and mathematical analysis [9]. However, these methods still rely heavily on expert and specialized tools. In practice, chip manufacturers and certification laboratories often need to conduct preliminary security assessments before formal evaluations. However, qualified SPA experts are scarce, and manual analysis typically requires several hours per trace, making it impractical to evaluate large numbers of implementations efficiently. This creates an urgent demand for automated SPA tools that can be operated by non-experts, enabling rapid preliminary assessments without the need for extensive domain knowledge or specialized equipment. As illustrated in Fig. 1, this work explores the potential of LLMs to bridge this gap, enabling non-experts to perform fully automated SPA on public-key cryptosystems through a single prompt interaction.

Our preliminary experiments reveal that directly applying LLMs to SPA tasks fails entirely—the models cannot correctly interpret power traces or recover secret keys without expert-level guidance. To address this challenge, we propose a novel prompt framework specifically designed for SPA, comprising six components including three expert strategies. This framework transforms the complex SPA task into a structured workflow that LLMs can effectively follow, enabling successful key recovery without requiring domain expertise from users. Our main contributions are as follows:

- We propose a novel prompt framework consisting of

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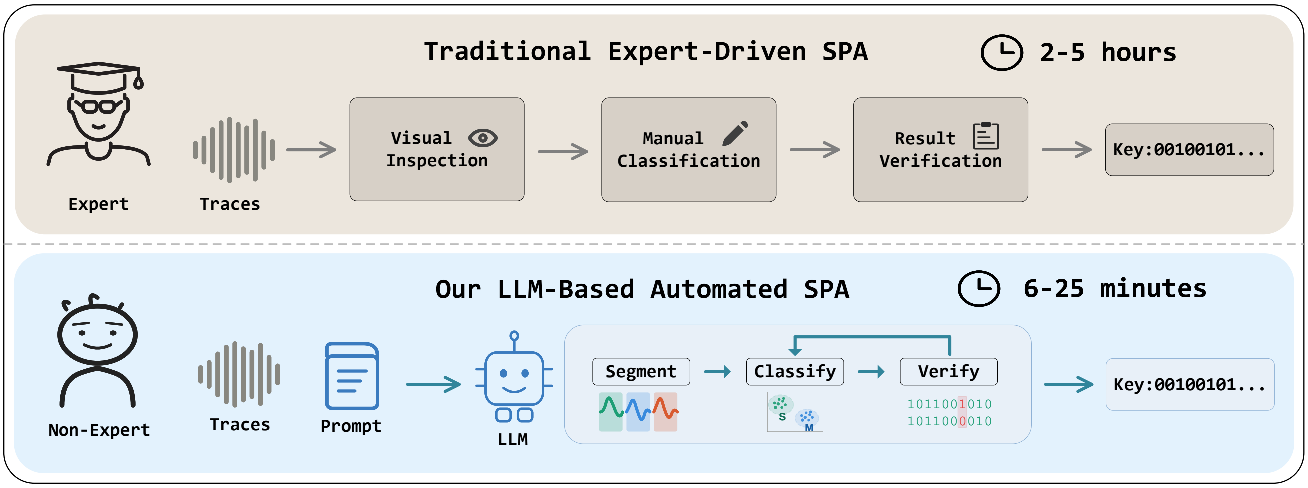


Fig. 1. Comparison between traditional expert-driven SPA and our LLM-based automated SPA.

six components: role, reinforce, input, task description, chain-of-thought (CoT), and expert strategies, with three expert strategies specifically designed for SPA tasks.

- We establish the first dataset for evaluating LLMs on SPA tasks, comprising twenty-five power traces from real cryptographic implementations and simulated leakage patterns, covering RSA, ECC, and Kyber.
- We conduct the first systematic evaluation of LLMs for SPA tasks. We evaluate two state-of-the-art LLMs, GPT-4o and DeepSeek-V3.1, and compare against a representative unsupervised horizontal clustering baseline. The baseline achieves only 52.04% accuracy, while our framework enables GPT-4o and DeepSeek-V3.1 to achieve overall average accuracies of 98.02% and 97.83%, respectively, with successful key recovery on all twenty-five traces. Additionally, our framework reduces analysis time by over 93.09% at comparable accuracy, enabling non-experts to complete SPA in minutes rather than hours.

The organization of this paper is as follows: Section II gives the background about SPA and LLM. Section III describes the research pipeline and expert strategies design. Section IV shows the experimental setup and evaluation results. Section V discusses the bottlenecks and future directions. Section VI concludes the paper.

II. Background

A. SPA on Public-key Cryptosystems

Side-channel analysis (SCA) exploits physical leakage from cryptographic devices, such as power consumption and electromagnetic emanations, to extract secret information. Among various SCA techniques, SPA recovers secret keys by directly inspecting power traces, without requiring statistical analysis over multiple measurements as in differential power analysis (DPA) [6]. This makes SPA particularly threatening in scenarios where the attacker can obtain only a single or very few traces.

Currently, RSA [10] and ECC [11], [12] are two widely deployed public-key cryptosystems. The sequence of cryptographic operations is directly related to the private key [13]–[15]. Taking RSA as an example, if a private key bit is “0”, the algorithm only performs modular squaring. However, if a private key bit is “1”, the algorithm sequentially performs modular squaring and modular multiplication. Due to the distinct operations for different bit values, these variations are often significantly reflected in power traces [16], [17].

As quantum computing technology continues to progress, RSA and ECC face potential threats from quantum attacks. To address this concern, NIST initiated a selection process for post-quantum cryptography, and Kyber has emerged as the first standardized post-quantum KEM algorithm. While Kyber exhibits robust algorithmic security, its implementation on certain physical devices remains vulnerable to side-channel leakage, particularly through SPA. SPA enables the classification of each secret bit as either 0 or 1, which is critical for extracting the secret key [18]. Therefore, the core of SPA lies in classifying power traces corresponding to different cryptographic operations, thereby recovering the private key.

To automate SPA, existing approaches typically adopt unsupervised learning techniques. A representative method proposed by Specht et al. [19] applies principal component analysis (PCA) as a preprocessing step before clustering, with a strategy to select few mid-ranked components where exploitable, low-variance leakage is concentrated, discarding both high-ranked components dominated by noise and low-ranked ones containing little useful information. The selected components are then classified via expectation-maximization or K-means clustering. This PCA-based horizontal clustering attack (HCA) has demonstrated significant improvements in non-profiled single-execution attacks on exponentiations. However, such methods rely on handcrafted feature extraction pipelines, and the optimal component selection is highly device- and application-specific, requiring manual parameter tuning that limits their generalizability across diverse targets.

B. LLMs and Prompt Engineering

LLMs have demonstrated powerful capabilities in language understanding and generation. Representative models include

OpenAI's GPT series [1] and the open-source DeepSeek models [2]. Users can interact with these models through web interfaces or APIs. LLMs have been successfully applied to various domains, including code generation [3], vulnerability management [4], and anomaly detection [5], enabling automated completion of complex tasks.

A common method to adapt general models to specific tasks is fine-tuning. However, fine-tuning is often impractical for domain-specific applications due to the need for large annotated datasets and significant computational resources. As a result, the success of LLMs in various domains largely relies on prompt engineering, i.e., optimizing the input prompts to improve the relevance and accuracy of model outputs [20]. Prompt engineering encompasses various strategies to enhance LLM performance without modifying model parameters. Basic approaches include zero-shot prompting, which directly describes the task, and few-shot prompting, which provides demonstration examples [21]. More advanced techniques incorporate supplemental information such as role definitions [22], [23] and step-by-step reasoning guidance [24]. These strategies have shown effectiveness across multiple domains [20], yet their potential for side-channel analysis tasks remains unexplored, motivating this work.

III. Prompt Framework for SPA

A. Prompt Framework

To enable non-experts to perform fully automated SPA using LLMs, we propose a prompt framework that embeds domain-specific expert knowledge into structured prompts. The overall research pipeline, as shown in Fig. 2, consists of three phases: ① framework design and dataset preparation, ② expert strategies design and optimization, and ③ systematic evaluation.

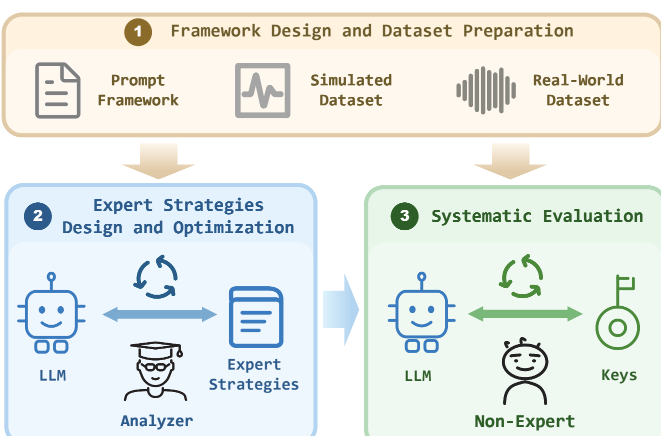


Fig. 2. Three-Phase Pipeline of Our Method.

In phase ①, we design the prompt framework based on widely adopted prompt engineering strategies [21]. The framework defines six components, as detailed in Table I. Fig. 3 provides a concrete instantiation of the framework for SPA tasks; users can adapt the framework to other side-channel analysis scenarios by modifying the expert strategies while retaining the overall structure. The first five components are derived from established prompt engineering techniques: *Role* assigns the model as a domain expert, *Reinforce* emphasizes key behavioral constraints to ensure the model follows the expected analysis workflow, *Input* specifies the trace data and its format, *Task Description* defines the classification objective, and *CoT* guides step-by-step reasoning. The sixth component, *Expert Strategies*, is specifically designed for SPA tasks and will be detailed in Section III-B. Additionally, we establish two datasets: a simulated dataset comprising eighteen sets of traces that cover typical SPA leakage patterns (Table II), and a real-world dataset comprising seven sets of traces from actual cryptographic implementations including RSA, ECC, and Kyber (Table III).

TABLE I
Components of the proposed prompt framework.

Component	Description
Role	Assign the model to act as a domain expert for the target task.
Reinforce	Emphasize key instructions to ensure the model follows the expected behavior.
Input	Specify the input data source and format.
Task Description	Define the task objective and expected output format.
CoT	Guide the model to reason step by step before providing the final answer.
Expert Strategies	Provide domain-specific knowledge and strategies tailored to the target task.

In phase ②, we design and optimize the expert strategies based on the standard workflow of human experts performing SPA. We first derive three initial strategies: **Classification**, **Constraint Verification**, and **Adaptive Retry**. We then iteratively refine these strategies using the simulated dataset, analyzing the LLM's intermediate reasoning process and adjusting the strategy formulations to improve accuracy. Details regarding the design and optimization process are discussed in Section III-B.

In phase ③, we validate the effectiveness and generalizability of our framework through systematic evaluation. We evaluate two state-of-the-art LLMs, GPT-4o and DeepSeek-V3.1, and compare the results against the horizontal clustering baseline described in Section II. For each trace, we conduct ten independent trials for both our approach and the clustering baseline, and compute the classification accuracy by comparing the output key sequence against the ground truth.

B. Expert Strategies Design and Optimization

The expert strategies are derived from the standard workflow of human experts performing SPA. When analyzing power traces, experts typically follow these steps: (1) segment the trace according to the known bit length of the private key [9], [25]; (2) classify the trace segments into two categories using methods such as visual inspection or clustering [19], [26], [27]; (3) verify the results against known constraints and retry with different methods if violations occur [28]. Finally, the private

	I would like you to act as an expert in side-channel analysis
Role	and signal processing, helping me analyze vulnerabilities in an
	encryption system. During the analysis process, please do not ask
Reinforce	me any questions; instead, proceed step by step until you provide
	the final answer. Classification labels must come from clustering
	output only. Do not fabricate any results without running actual
	code. The file contains a power consumption trace from an ECC
Input	signing process, where each segment represents an operation.
Task	There are 288 operations in this waveform. Please identify these
Description	operations, which can be classified into two types: D and A.
	To complete this task, you need to follow these steps:
	1. Segment the trace into exactly 288 equal-length parts.
Zero-CoT	2. Classify each operation into D and A. Apply dimensionality
	reduction (e.g., PCA), then apply clustering (e.g., K-means with
	k=2) to classify the operations into two categories. Assign the
	larger cluster as D.
	3. After every classification attempt, verify: exactly 192 D
Expert	operations, first operation is D, and no two consecutive A
Strategies	operations. Print the verification results.
	4. If constraints are not satisfied, adjust the dimensionality
	reduction parameters or switch to alternative methods (e.g., t-
	SNE), and try different clustering algorithms (e.g., GMM,
	DBSCAN). Keep iterating until all constraints are met.

Fig. 3. An instantiation of the proposed prompt framework for ECC-based SPA.

key is reconstructed based on the classified operations [13]–[15].

In this work, we focus on scenarios where segmentation can be performed straightforwardly (e.g., equal-length partitioning based on known key length), and incorporate the segmentation instruction into the CoT component of the prompt framework. The discussion of complex segmentation scenarios is deferred to Section V. Based on steps (2)-(3) of the expert workflow, we formulate three expert strategies:

- **Classification**: Instruct the model to apply dimensionality reduction followed by clustering to separate trace segments into two operation categories.
- **Constraint Verification**: Encode domain-specific constraints that enable the model to verify its classification results, such as the expected number of each operation type and their ordering rules.
- **Adaptive Retry**: Guide the model to systematically adjust its analysis when constraints are not satisfied, such as modifying dimensionality reduction parameters or switching to alternative algorithms, and re-execute the classification until all constraints are met.

These three strategies form a closed loop, where the classification-verification-retry cycle continues until all domain constraints are satisfied.

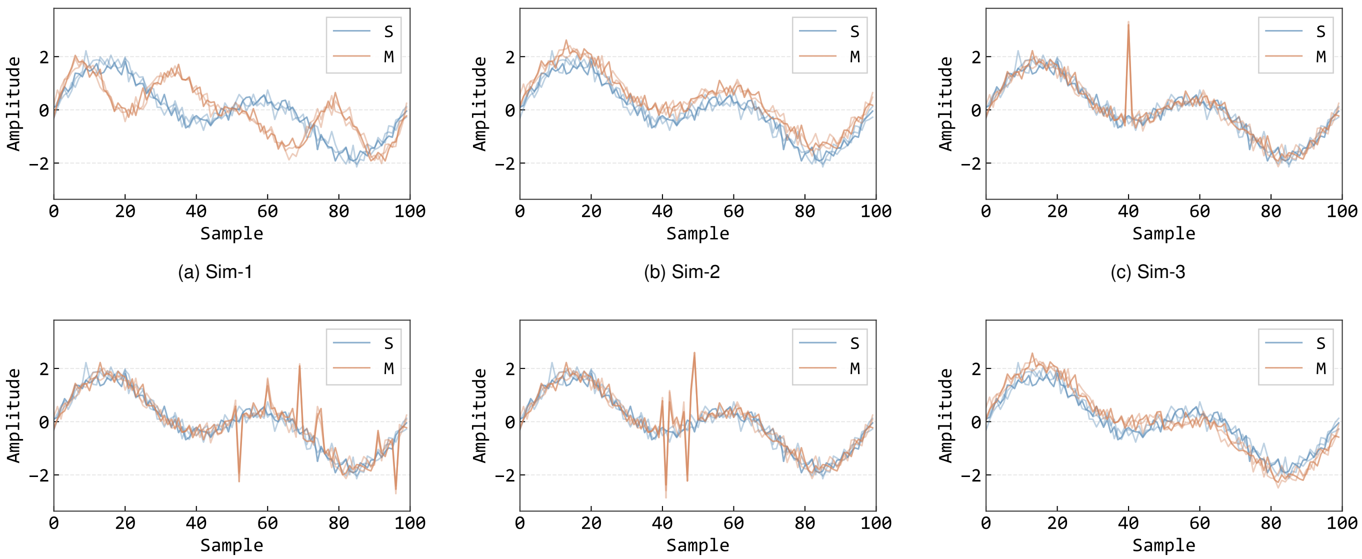
To ensure the effectiveness and robustness of these strategies, we conducted iterative refinement using the simulated dataset. Specifically, we analyzed the LLM's intermediate reasoning process to identify which step the model frequently failed at, be it classification, constraint verification, or adaptive retry, and then injected targeted expert guidance into the corresponding strategy to address the failure pattern. This refinement process was repeated until the accuracy on the simulated dataset stabilized. Fig. 3 illustrates a complete prompt instantiation for SPA on an ECC implementation, where the operations D and A denote point doubling and point addition, respectively. The expert strategies are highlighted in pink. The framework is designed to be generalizable: adapting it to other public-key algorithms requires modifications primarily to the expert strategies, such as updating the algorithm-specific operation descriptions and constraint rules (e.g., replacing the 288 operations and 192 D operations in the ECC example with the corresponding values for the target algorithm), while the remaining five components can be reused directly.

IV. Evaluation Experiments

In this section, we present the evaluation experiments to validate the effectiveness of our proposed prompt framework for SPA tasks. We first describe the experimental setup, including the datasets and evaluation metrics. Then, we compare our framework against the horizontal clustering baseline and evaluate LLM performance on two state-of-the-art models: GPT-4o and DeepSeek-V3.1, to demonstrate the framework's effectiveness and generalizability.

A. Experimental Setup

LLM Configuration. We evaluate two state-of-the-art LLMs: GPT-4o (snapshot: gpt-4o-2024-11-20) and DeepSeek-V3.1-Terminus. We build a unified conversational platform using LangChain [29], equipped with a Python code interpreter as an external tool, enabling LLMs to autonomously write and execute analysis scripts during the SPA process. We access GPT-4o through OpenRouter's API [30] and DeepSeek-V3.1-Terminus through SiliconFlow's API [31]. For both models,



(a) Sim-1 (b) Sim-2 (c) Sim-3

(d) Sim-4 (e) Sim-5 (f) Sim-6

Fig. 4. Simulated trace patterns under σ_{20} noise.

each experiment starts a new conversation to ensure independent trials.

Evaluation Metric. For each trace, we conduct ten independent trials for both our approach and the clustering baseline, with each trial requiring only a single prompt input. We measure accuracy as the proportion of correctly recovered key bits across these trials.

Dataset. We establish a dataset[1] comprising a simulated dataset and a real-world dataset. The simulated dataset is designed to cover a wide range of typical SPA leakage scenarios, which we use to iteratively refine our expert strategies. The real-world dataset contains traces from actual cryptographic implementations, providing validation on practical scenarios.

The simulated dataset includes six distinct categories (Sim-1 through Sim-6), each with three noise levels using Gaussian noise with standard deviations of 0.01, 0.1, and 0.2, denoted as σ_1, σ_{10}, and σ_{20}, respectively. Using RSA's modular square (S) and modular multiplication (M) operations as examples, we characterize the difference between two operation traces by Euclidean distance:

$$d_{\text{Euclidean}} = \sqrt{\sum_{t=1}^{N} (y_1(t) - y_2(t))^2} \quad (1)$$

where t denotes the t-th point, N represents the total number of points in each operation, and $y_1(t)$, $y_2(t)$ are the trace values at point t. The six categories are designed with comparable Euclidean distance but different leakage patterns, as summarized in Table II. All categories simulate a 1024-bit key with 1536 operations, where each operation consists of 100 points. Fig. 4 visualizes each category under σ_{20} noise, where three operation cycles of S (blue) and M (orange) are overlaid to illustrate the leakage patterns.

[1]https://github.com/haillife/One-Solves-All

TABLE II
OVERVIEW OF THE SIMULATED DATASET.

Name	Ops	Leakage Pattern
Sim-1	1536	S and M have distinct overall waveform shapes
Sim-2	1536	M has slightly higher amplitude than S
Sim-3	1536	M differs from S at a single sample point
Sim-4	1536	M has 10 discrete points differing from S
Sim-5	1536	M has 10 consecutive points differing from S
Sim-6	1536	S and M have symmetric amplitude differences

The real-world dataset includes seven sets of traces from different cryptographic implementations: four RSA implementations on a contract card, ASIC, FPGA, and STM32F429, two ECC implementations on an AT89S52 and a contract card, and one post-quantum Kyber implementation on an STM32F407 [18]. Table III presents the details including private-key lengths, number of operations, total trace points, trace types, devices, and implementation types. Since this work focuses on the classification aspect of SPA rather than segmentation, three traces (RSA on ASIC, ECC on AT89S52, and Kyber on STM32F407) were preprocessed by simply padding or truncating each operation to a uniform length, ensuring straightforward equal-length partitioning for SPA evaluation.

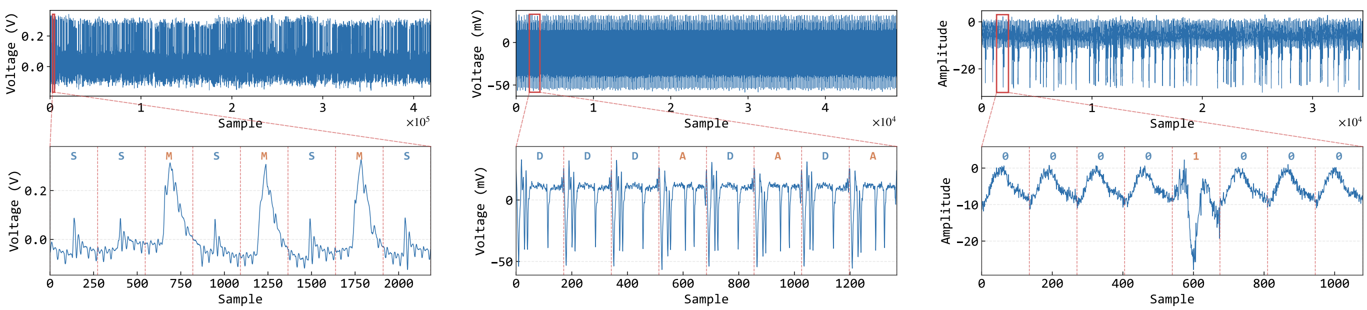
Fig. 5 shows three representative real-world traces from the dataset. Each trace is displayed with the full trace (top) and a zoomed-in view (bottom) highlighting the operation boundaries and classification labels.

B. Experimental Results

In this section, we evaluate the proposed framework from four perspectives: overall effectiveness, representative case

TABLE III
OVERVIEW OF THE REAL-WORLD DATASET.

Device	Algorithm	L_{key}	Operations	Points	Points/Op	Bit-1 Ratio	Trace Type	Implementation
Contract card	RSA	1024	1561	451129	289	52.4%	Power	co-design
ASIC	RSA	1024	1536	502272	327	50.0%	Power	hardware
FPGA	RSA	1024	1536	2443776	1591	50.0%	Power	hardware
STM32F429	RSA	1024	1534	418782	273	50.1%	Power	software
AT89S52	ECC	192	288	49248	171	50.0%	Power	software
Contract card	ECC	256	384	106752	278	50.0%	Power	co-design
STM32F407	Kyber	256	256	34560	135	—	EM	software



(a) RSA on STM32F429 (b) ECC on AT89S52 (c) Kyber on STM32F407

Fig. 5. Representative real-world traces with zoomed-in views.

studies, robustness, and time efficiency. We compare our method with HCA based on dimensionality reduction and clustering, and analyze how the framework supports automated iterative classification on both real-world and simulated traces.

1) Overall Effectiveness: We evaluate the proposed prompt framework by comparing it against the PCA-based HCA [19] described in Section II. Following the original paper, we apply PCA to the trace-segment matrix and select a single mid-ranked principal component (the fourth) for K-means clustering, which corresponds to the experimentally optimal configuration reported therein. We test two state-of-the-art LLMs, GPT-4o and DeepSeek-V3.1, with our framework on all twenty-five traces. Table IV presents the classification accuracy for all three approaches.

Fig. 6 illustrates a complete analysis case under our framework. The user provides a prompt together with the trace files as input, and the LLM then performs the analysis by following the guidance of our framework. Specifically, the framework guides the model to carry out the analysis in a structured manner, including segmentation, classification, and rule verification. If the verification is successful, the model directly outputs the recovered key sequence. Otherwise, the framework further guides the model to adjust parameters or switch methods, and then repeat the classification and verification process. This iterative loop continues until the result satisfies the predefined constraints, after which the final key sequence is returned. The example demonstrates that the main contribution of our framework is not merely to provide a prompt, but to organize the LLM's behavior into a closed-loop analysis procedure with verification and self-correction, thereby improving the reliability and automation of SPA.

The results in Table IV and Fig. 7 clearly demonstrate the effectiveness of our framework. HCA achieves only 52.04% accuracy across all traces, which is close to random guessing for a binary classification task. In contrast, with our proposed framework, GPT-4o and DeepSeek-V3.1 achieve overall average accuracies of 98.02% and 97.83%, respectively, with successful key recovery on all twenty-five traces.

The key reason for this performance gap lies in the adaptability of the two approaches. HCA follows a fixed pipeline with predetermined parameters, and without expert intervention, there is no mechanism to adjust the dimensionality reduction or clustering configuration when the initial attempt fails. In contrast, our framework embeds expert knowledge into the prompt, enabling the LLM to autonomously try different combinations of dimensionality reduction methods and clustering algorithms, and iteratively adjust parameters until the domain-specific constraints are satisfied. This effectively achieves automated expert-level analysis without requiring actual expert participation.

Both models achieve slightly higher accuracy on the simulated dataset (98.67% and 98.08%) than on the real-world dataset (96.35% and 97.20%), though the difference is marginal. This is primarily because the simulated traces are generated with naturally aligned leakage points in the time domain, making them inherently easier to classify. Real-world traces, in contrast, cannot be perfectly aligned due to hardware variations, introducing minor additional difficulty.

TABLE IV
CLASSIFICATION ACCURACY (%) ON ALL TWENTY-FIVE TRACES.

Trace	Config.	HCA	GPT-4o	DeepSeek
Contract card	RSA	51.12	94.84	90.23
ASIC	RSA	50.07	100	100
FPGA	RSA	50.07	95.02	100
STM32F429	RSA	51.43	100	100
AT89S52	ECC	52.08	100	95.46
Contract card	ECC	53.39	89.47	94.71
STM32F407	Kyber	65.23	95.12	100
Sim-1	σ_1	50.20	100	100
	σ_{10}	51.95	100	100
	σ_{20}	51.63	100	95.38
Sim-2	σ_1	50.39	100	100
	σ_{10}	50.72	100	100
	σ_{20}	50.00	95.31	100
Sim-3	σ_1	55.34	100	95.84
	σ_{10}	55.66	94.66	100
	σ_{20}	55.73	100	94.27
Sim-4	σ_1	50.46	100	100
	σ_{10}	50.78	100	100
	σ_{20}	50.26	95.78	89.91
Sim-5	σ_1	50.65	100	100
	σ_{10}	50.13	100	95.62
	σ_{20}	50.91	94.53	100
Sim-6	σ_1	51.76	100	100
	σ_{10}	50.13	95.84	100
	σ_{20}	50.85	100	94.38
Average		52.04	**98.02**	**97.83**

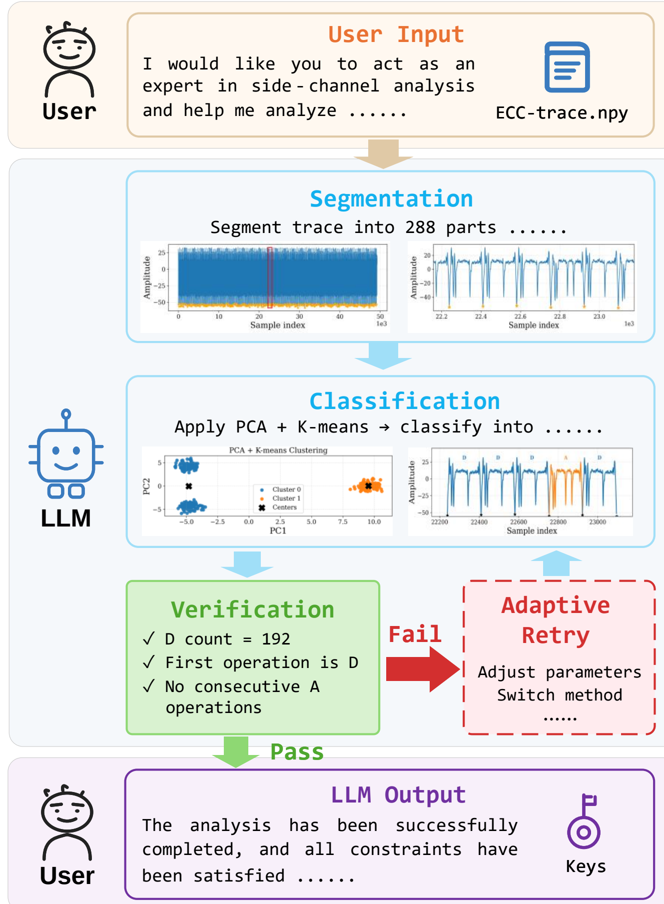


Fig. 6. A complete experimental process from prompt input to key recovery output.

Nevertheless, the high accuracy on both datasets confirms the framework's practical applicability.

Notably, GPT-4o and DeepSeek-V3.1 achieve comparable performance overall, with neither model consistently outperforming the other across all traces. This suggests that our framework generalizes well across different LLMs, and the quality of the expert strategies matters more than the choice of a specific model.

2) Case Studies: To provide a more intuitive understanding of how the proposed framework outperforms conventional HCA on real-world traces, we present two representative case studies highlighting different failure modes. Note that unlike the fixed HCA configuration used in Table IV, the following case studies explore HCA with trace-specific configurations to demonstrate that even with more favorable parameters, HCA still fails without an iterative adjustment mechanism. In contrast, our framework can automatically revise the analysis configuration through iterative retry until the domain constraints are satisfied, effectively replicating the trial-and-error process of a human expert.

a) RSA on Smart Card: We present a detailed case study on the RSA implementation running on a smart card. Fig. 8 illustrates the classification results from both HCA and our framework across multiple iterations.

HCA directly applies PCA to the full operation segments and then performs K-means clustering with PC2 and PC3, yielding a classification accuracy of approximately 61.8% (Fig. 8(a)). The main limitation of this method lies in the quality of the reduced feature space. Although PCA compresses the original waveform into a lower-dimensional representation, the resulting PC2–PC3 projection still does not provide sufficiently clear separability between S and M operations. Consequently, the subsequent clustering stage cannot produce a correct partition. In traditional analysis, such a failure would typically require human experts to manually inspect the projection and repeatedly adjust the dimensionality configuration, making the process labor-intensive and expertise-dependent.

When our framework is applied, the LLM also fails in its first iteration (Fig. 8(b)), and in fact performs worse than HCA. Its initial dimensionality reduction configuration produces an even poorer projection space, so the two operation types are mixed more severely and clustering accuracy drops to approximately 51.1%. Importantly, however, the framework can automatically identify this failure through *Constraint Verification*, and then invoke *Adaptive Retry* to continue revising the analysis. In the second iteration (Fig. 8(c)), the LLM obtains a noticeably improved low-dimensional representation, indicating that the dimensionality reduction step has moved in the right direction. Even so, the clustering result is still

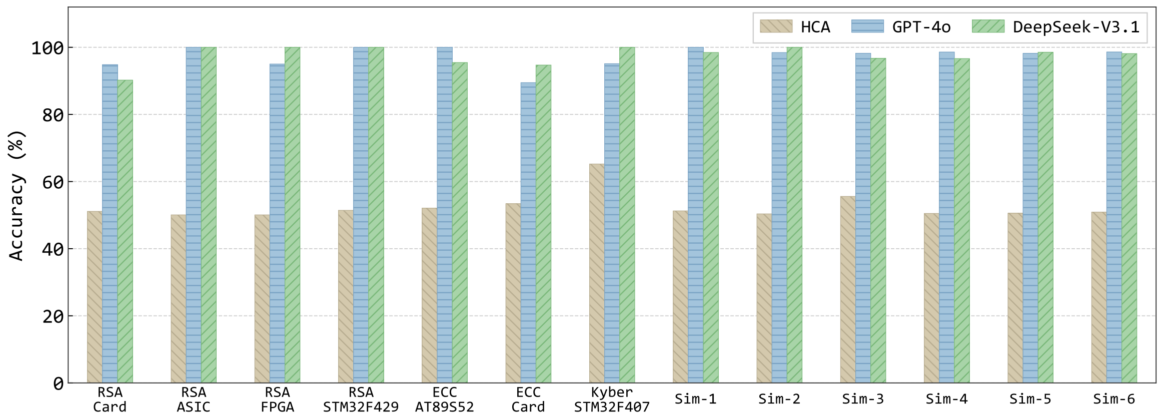
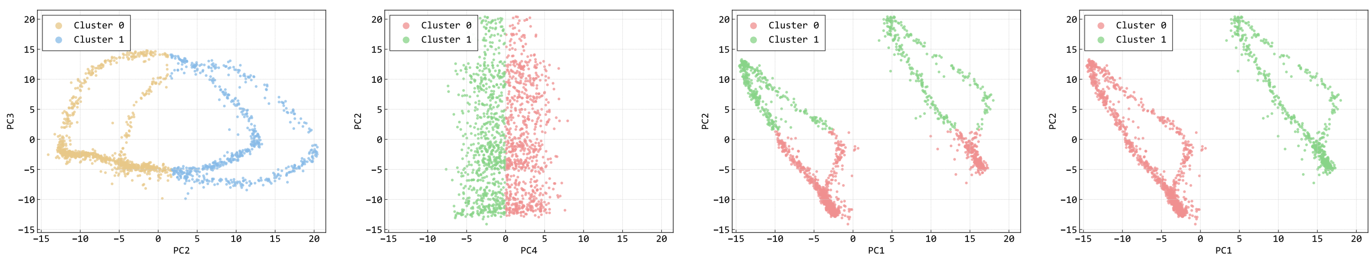
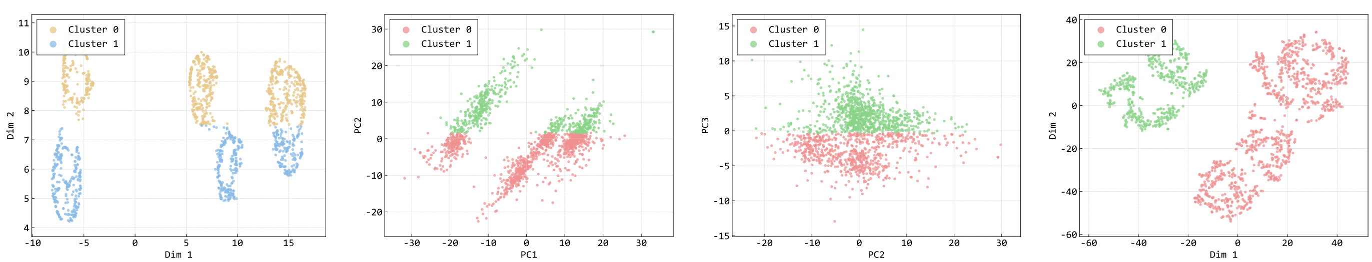


Fig. 7. Accuracy comparison across all traces. The left seven groups correspond to the real-world dataset and the right six groups correspond to the simulated dataset (averaged across three noise levels).



(a) HCA baseline (61.8%) (b) LLM: Iteration 1 (51.1%) (c) LLM: Iteration 2 (61.7%) (d) LLM: Iteration 3 (100%)

Fig. 8. Case study: RSA on smart card. HCA provides a limited baseline, while our framework progressively improves the classification across three iterations and finally achieves 100.0% accuracy.



(a) HCA baseline (56.5%) (b) LLM: Iteration 1 (71.5%) (c) LLM: Iteration 2 (60.8%) (d) LLM: Iteration 3 (100%)

Fig. 9. Case study: RSA on STM32F429. HCA provides a limited baseline, while our framework iteratively revises the analysis configuration and finally achieves near-perfect classification.

incorrect, meaning that a better projection alone is not yet sufficient for fully correct classification. Upon entering the third iteration (Fig. 8(d)), the framework further adjusts the clustering method and parameters, and eventually finds a valid classification that satisfies all domain-specific constraints, with the accuracy reaching 100%.

b) RSA on STM32F429: We further present a detailed case study on the RSA implementation running on an STM32F429. Fig. 9 shows the classification results from HCA and from our framework across three iterations.

HCA first applies UMAP to obtain a two-dimensional representation and then performs GMM clustering, achieving a classification accuracy of 56.5% (Fig. 9(a)). In this case, the main limitation is not that the low-dimensional projection is completely unusable. On the contrary, the reduced space already exhibits a relatively clear structure. However, the subsequent clustering configuration fails to identify the correct partition of S and M operations, leading to incorrect classification. In traditional practice, this kind of failure would still require a human expert to manually inspect the projected distribution and revise the clustering strategy.

When our framework is applied, the LLM does not solve the problem immediately, but its first iteration (Fig. 9(b)) already produces a reasonably good low-dimensional representation. The remaining error mainly lies in the clustering stage rather than in dimensionality reduction itself, and the accuracy improves to 71.5%. Through *Constraint Verification*, the framework can automatically detect that the obtained result

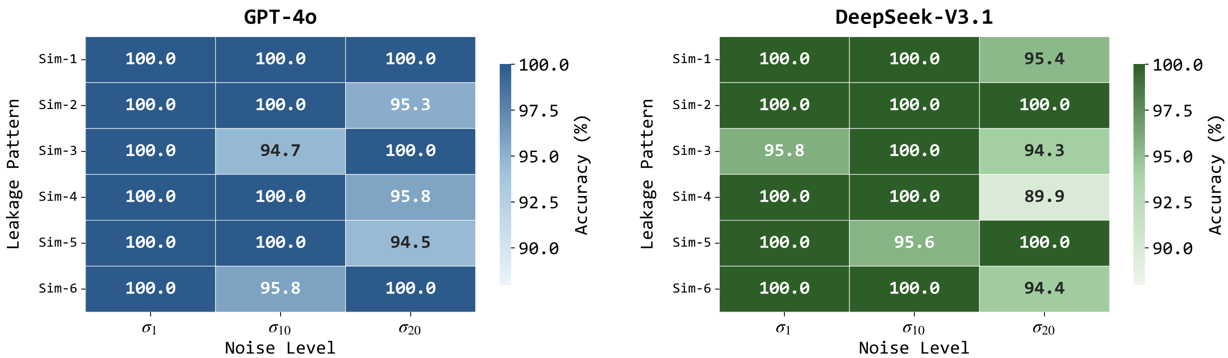


Fig. 10. Accuracy heatmap on the simulated dataset across leakage patterns and noise levels.

is still inconsistent with the domain constraints, and *Adaptive Retry* is then triggered to revise the analysis configuration. In the second iteration (Fig. 9(c)), the LLM explores another parameter configuration, but this attempt moves in the wrong direction: the dimensionality reduction quality deteriorates, which in turn makes the clustering result worse, and the accuracy drops to 60.8%. This intermediate failure is important, because it shows that the framework does not rely on a single lucky attempt, but can continue exploring alternative configurations even after a poorer intermediate result. When the framework enters the third iteration (Fig. 9(d)), it further revises the clustering configuration, and ultimately obtains a result that satisfies all domain-specific constraints, achieving 100.0% classification accuracy.

These cases clearly demonstrate the core advantage of our framework over HCA: the ability to autonomously iterate through different parameter configurations until domain constraints are satisfied, effectively replicating the trial-and-error process of a human expert.

3) Robustness Analysis: We further analyze the framework's robustness with respect to noise levels and leakage patterns using the simulated dataset. Fig. 10 presents the accuracy heatmap for GPT-4o and DeepSeek-V3.1 across six leakage patterns and three noise levels.

Regarding noise robustness, as the noise level increases from σ_1 to σ_{20}, no systematic accuracy degradation is observed for either model. For example, both GPT-4o and DeepSeek-V3.1 achieve 100% accuracy on multiple traces even at the highest noise level σ_{20}. Given that each accuracy value is computed from ten independent trials, minor fluctuations (e.g., between 95% and 100%) are within the expected statistical variance. The absence of a consistent downward trend indicates that the framework's Adaptive Retry mechanism effectively compensates for noise-induced classification errors by automatically adjusting dimensionality reduction parameters or switching to alternative algorithms.

Regarding leakage pattern robustness, all six simulated categories achieve high accuracy without any category exhibiting consistently poor performance. The lowest individual result is 89.91% (Sim-4 with σ_{20} on DeepSeek-V3.1), which corresponds to the combination of the highest noise level and discrete multi-point leakage — the most challenging scenario where subtle features are scattered across the trace. Nevertheless, this still represents a substantial improvement over the baseline's approximately 50% accuracy. Overall, the framework demonstrates strong adaptability across diverse leakage patterns.

4) Time Efficiency Comparison: Table V compares the time consumption between manual analysis by human experts and LLM-based analysis using our framework. Manual SPA requires domain expertise and typically takes 120–300 minutes per trace, involving steps such as visual inspection, pattern recognition, and result verification. In contrast, our framework enables fully automated analysis in 6–25 minutes per trace, requiring only a single prompt input without any expert intervention. The LLM time reported is the average across GPT-4o and DeepSeek-V3.1. On average, our framework reduces the analysis time by 93.09%. Moreover, unlike human experts who are scarce, expensive, and limited by working hours, LLMs can operate continuously around the clock, making it feasible to evaluate large batches of cryptographic implementations — a scenario that would be prohibitively costly with manual analysis.

TABLE V
TIME CONSUMPTION COMPARISON BETWEEN MANUAL ANALYSIS AND LLM-BASED ANALYSIS.

Device	Algorithm	Manual (min)	LLM (min)	Reduction (%)
Contract card	RSA	285	22	92.28
ASIC	RSA	256	18	92.97
FPGA	RSA	294	25	91.50
STM32F429	RSA	237	15	93.67
AT89S52	ECC	153	8	94.77
Contract card	ECC	178	12	93.26
STM32F407	Kyber	124	6	95.16
Average		218.14	15.14	**93.09**

V. Discussion

In this section, we reflect on the design rationale of our framework, discuss the scope and limitations of the expert strategies, and outline promising directions for future research.

A. High-level Guidance vs. Rigid Specifications

Through our experiments, we observed that LLMs benefit more from high-level guidance than from rigid, detailed specifications. For instance, our Classification strategy suggests candidate methods such as “apply dimensionality reduction (e.g., PCA), then apply clustering (e.g., K-means),” rather than mandating exact hyperparameters or a fixed pipeline. This design choice is intentional: LLMs excel at interpreting analytical intent and generating appropriate implementations, but may become confused or constrained when given overly prescriptive instructions.

This observation is further supported by our finding that minor variations in prompt wording have negligible impact on performance. For example, using “trace” versus “waveform” to describe the input data yields nearly identical results. Similarly, describing the classification task as “divide into two categories” versus “classify into S and M operations” does not significantly affect accuracy. This suggests that LLMs are robust to surface-level linguistic variations and instead focus on the underlying semantic meaning of the instructions.

These findings have important implications for prompt engineering in domain-specific tasks: practitioners should focus on conveying clear analytical strategies and domain constraints, rather than attempting to specify every procedural detail.

B. Scope and Limitations of Expert Strategies

As discussed in Section III-B, our three expert strategies are derived from the standard workflow of human experts performing SPA. These strategies cover the core steps of the analysis process, while trace segmentation is incorporated into the CoT component and assumed to be straightforward (e.g., equal-length partitioning based on known key length).

However, segmentation can become complex in certain scenarios, such as when operations have variable execution times or when the trace contains significant timing jitter. In such cases, a promising direction is to apply the same methodology used in this work: encoding existing segmentation techniques as additional expert strategies and injecting them into the prompt framework, enabling the LLM to autonomously select and apply appropriate segmentation methods before proceeding to classification.

Additionally, we note that the Constraint Verification strategy requires adaptation for different algorithms. For example, in RSA implementations, the constraints include rules such as “there must be at least one S operation between any two M operations,” which reflect the structure of the square-and-multiply algorithm. However, in Kyber implementations where the goal is to distinguish between consecutive bits rather than between distinct operation types, the constraint rules need to be reformulated accordingly. Users should adapt the specific constraints based on the characteristics of the target algorithm, while the overall three-strategy loop remains applicable.

C. Future Directions

Several promising directions emerge from this work. First, while we evaluated GPT-4o and DeepSeek-V3.1, extending the evaluation to other LLMs would provide a more comprehensive understanding of which model capabilities are most critical for SPA tasks. Second, exploring more complex SPA scenarios, such as protected implementations with countermeasures or multi-trace analysis, would test the limits of our framework. Third, investigating automatic prompt optimization techniques could reduce the manual effort required to design effective expert strategies. Finally, combining LLM-based analysis with traditional machine learning methods may yield complementary strengths and further improve accuracy.

VI. Conclusion

In this paper, we conduct the first systematic evaluation of LLMs for SPA tasks. We propose a novel prompt framework consisting of six components (role, reinforce, input, task description, chain-of-thought, and expert strategies), with three expert strategies specifically designed for SPA: Classification, Constraint Verification, and Adaptive Retry. To validate the framework, we establish a dataset comprising eighteen simulated traces covering typical SPA leakage patterns and seven real-world traces from various cryptographic implementations including RSA, ECC, and Kyber.

Our experimental results on GPT-4o and DeepSeek-V3.1 demonstrate that the proposed framework significantly outperforms the HCA baseline, which achieves only approximately 52% accuracy. In contrast, our framework enables both models to successfully recover private keys across all twenty-five traces, achieving overall average accuracies of 98.02% and 97.83%, respectively. Furthermore, our framework reduces analysis time by over 93%, from hours to minutes, enabling non-experts to perform fully automated SPA with minimal effort.

This work demonstrates the potential of LLMs as powerful tools for side-channel analysis when guided by well-designed prompts. Future research directions include extending the framework to more complex SPA scenarios, exploring automatic prompt optimization, and investigating LLM-assisted trace segmentation.

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